Limits of structure-conditioned active learning for sparse alchemical hydration calculations

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Abstract

Accurate hydration free energies require both a physical model and enough sampling of its alchemical response. Active SolvAI tested whether sparse response observations from a query molecule could update a structure-derived prior, reconstruct the same-Hamiltonian integral and improve prediction of experiment at lower measured cost. The answer was negative for the inherited 5-ps PIMD2 protocol. In a prospectively frozen 72-molecule endpoint test, actual-minus-prior response residuals increased five-partition mean absolute error from 0.186 to 0.190 kcal mol^{-1} and did not beat molecule-shuffled responses. We then generated 144 new windows for a prospectively frozen dense panel. At five and seven observed windows, molecule-conditioned Bayesian quadrature gave integral errors of 1.701 and 1.608 kcal mol^{-1}, compared with 1.153 and 1.092 for simple uniform direct integration. A non-deployable oracle appeared favorable when it selected and scored windows on the same short curve, but this advantage reversed under held-out trajectory blocks. These results distinguish apparent in-sample placement headroom from prospectively identifiable and statistically resolvable information.

1 Introduction

SolvAI established that structure-predicted solvent responses add information beyond matched molecular features and experimental labels. It did not establish that a high-fidelity alchemical response can be recovered from structure, that a few target-molecule observations contain endpoint information absent from the frozen predictor, or that posterior uncertainty can guide simulation. We therefore tested a distinct proposition: an actual response may reveal the molecule-specific departure from a structure-derived prior and indicate what to measure next.

We separated two targets that are often conflated. The first was the dense integral produced by the same short-window ARROW/PIMD2 Hamiltonian. The second was experimental hydration free energy, which additionally contains sampling error, Hamiltonian bias and experimental discrepancy. Improvement on one target was not allowed to stand in for improvement on the other. All molecule sets, response subsets, policies, budgets, controls and interpretation thresholds were fixed before their decisive outcomes were revealed.

2 Results

2.1 Actual-observation gate

The no-new-simulation gate used 72 ARROW molecules with complete PIMD2 observations at $\lambda = 0.1$, 0.5 and 0.9. Every endpoint correction was nested within molecule-held-out validation. Actual-minus-structure-predicted response residuals increased the five-partition mean absolute error from 0.1864 to 0.1899 kcal mol^{-1} (paired change +0.00348; 95% molecule-bootstrap interval

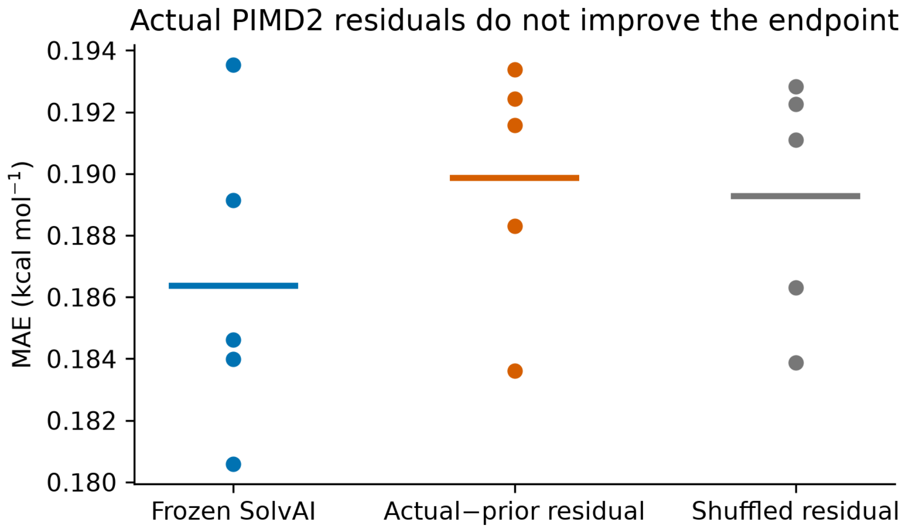

Figure 1: **Actual sparse PIMD2 observations do not improve the experimental endpoint.** Points are five independent molecule partitions; horizontal bars show their means. All models use identical held-out molecules.

+0.00096 to +0.00609). Only 31.9% of molecules improved, and the aligned residual was statistically indistinguishable from the mean of five molecule-shuffled controls. Raw actual responses, predicted responses and all predeclared one-, two- and three-point subsets likewise failed to improve the frozen endpoint.

2.2 Sparse reconstruction diagnostic

The endpoint route was killed by its prospective rule. A distinct reconstruction diagnostic remained: molecule-conditioned Gaussian posteriors reduced hidden-point errors relative to a generic population Gaussian for several predeclared subsets, with approximately nominal interval coverage. Absolute errors nevertheless remained at least 2.4 kcal mol^{-1} and 95% intervals were broad. Direct integration of the three actual observations gave 1.43 kcal mol^{-1} experimental MAE after fold-local affine calibration. Because no compatible dense historical population was available, these tests authorized only one bounded prospective dense sentinel, not a reconstruction or acceleration claim.

2.3 Prospective sentinels

We generated a common 15-point response grid for four calibration and eight prospective molecules. Each molecule reused three inherited windows and added twelve 5-ps PIMD2 windows. Calibration selected the Gaussian-process length scale and noise inflation before any prospective response was generated. All 96 prospective windows passed prespecified quality control on the first attempt.

The molecule-conditioned policy failed its frozen primary comparisons (Fig. 2). At five observed windows, active Bayesian quadrature had an integral MAE of 1.701 kcal mol^{-1}, compared with 1.153 for uniform direct integration; the paired difference was +0.548 kcal mol^{-1} (90% bootstrap interval −0.101 to +1.159), with improvement for two of eight molecules. At seven windows the corresponding errors were 1.608 and 1.092 kcal mol^{-1} (difference +0.516, interval −0.392 to +1.491),

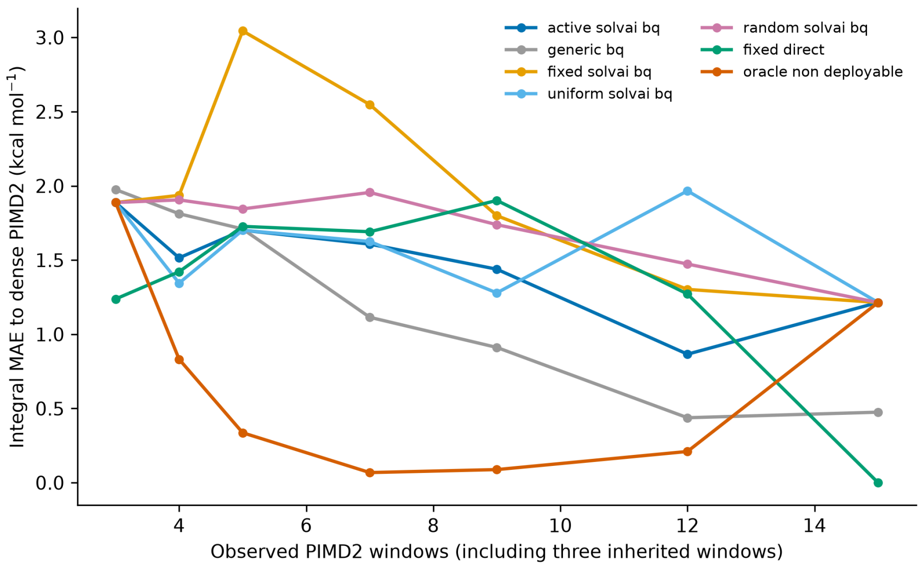

Figure 2: **Prospective same-Hamiltonian reconstruction does not favour active allocation.** Mean absolute error of the reconstructed 15-point short-window PIMD2 integral for eight prospective molecules. All methods receive the same inherited three windows and are compared at identical window counts. The oracle is non-deployable.

with three of eight improved. Coverage of the nominal 90% interval was 0.750 and 0.875, but mean interval widths remained 5.406 and 4.893 kcal mol^{-1}. No Bayesian method reached the frozen stopping threshold.

The same-curve non-deployable oracle reached 0.337 and 0.068 kcal mol^{-1} at five and seven windows, but it selected and evaluated windows against the same noisy 5-ps curve. A separately frozen independent-noise audit selected locations on non-overlapping blocks and scored them against held-out blocks. The oracle was then worse than uniform direct integration by 0.431 and 0.409 kcal mol^{-1} at five and seven windows, selected sets had reversal Jaccard similarities of 0.139 and 0.238, and complementary-half dense integrals differed by 1.866 kcal mol^{-1} on average. The original oracle therefore did not establish stable molecule-specific placement headroom.

3 Discussion

The program produced two linked null results. First, target-molecule PIMD2 measurements at the inherited three windows did not improve SolvAI's experimental prediction. Second, the structure-conditioned policy did not reduce the cost of dense reconstruction under the same Hamiltonian. The same-curve oracle initially suggested a gap between useful measurements and prospectively identifiable measurements, but independent-block scoring showed that the apparent gap was inseparable from finite-trajectory selection bias.

The 5-ps windows were noisy: the mean five-block standard error across the prospective dense panel was 2.283 kcal mol^{-1}, with a maximum of 15.087. Before launching the proposed longer-replica resolution experiment, we prospectively froze a power gate based on the observed noise scaling and molecule-level heterogeneity. For eight molecules with two 50-ps replicas at all 15 windows, simulated power to detect a 0.30-kcal mol^{-1} placement benefit was effectively zero at both registered budgets, while the probability of obtaining a reliable dense reference was 0.446. The proposed 12-ns PIMD2 campaign was therefore not run. Hamiltonian bias cannot explain failure against the same-Hamiltonian dense reference, although it may help explain why PIMD2 residuals do not track experimental error. Cross-fidelity allocation was deliberately not tested because the same-fidelity prerequisite failed.

The result sets a concrete threshold for future work. A larger protocol-matched dense-curve corpus would need to demonstrate accurate structure-to-curve prediction and prospective selection of informative windows before trajectory extension, PIMD4/PIMD8 escalation or a blind external campaign is warranted. Tier-B was not opened. The present evidence therefore supports a no-go decision for this Active SolvAI formulation, not a general claim that adaptive free-energy simulation is impossible.

4 Methods

The parent SolvAI artifact, cohorts, splits and response campaigns are versioned by SHA-256. The primary endpoint gate was committed before scoring at Git commit `a0dd986`. All preprocessing, response prediction, ridge selection and endpoint calibration occur inside molecule-held-out training data. Five repeated five-fold partitions use seeds 314159, 271828, 161803, 141421 and 173205. Destructive controls jointly permute response rows within the outer training and test folds, preserving the response distribution while removing molecule alignment. Paired intervals use 100,000 molecule bootstrap resamples with seed 20260828.

The dense-sentinel protocol was committed at `512aad3`; a prospective-noise leakage correction was committed before any sentinel response was generated, and numeric calibration was locked at `3e69d10`. Native Arbalest used force field 933, explicit water in a 24-Å periodic box, 298 K, 1 bar, two PIMD beads with PILE, a 0.002-ps step, 500 equilibration steps and 2,500 production steps per window. The fixed grid was $\lambda = \{0, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1\}$. The dense reference is the trapezoidal integral of these short-window means, not a converged PIMD8 free energy.

The prospective campaign generated 144 new windows: 48 calibration and 96 evaluation. All passed prespecified completeness, finiteness, temperature and density checks on the first attempt. Production totalled 720 ps, 288 bead-windows and 900,000 nominal bead-steps; measured GPU work was 1.932 GPU-h. Arbalest does not expose an exact fast/slow force-kernel call count, so nominal bead-steps are reported instead. Every attempted run, including failures, is retained in the append-only compute ledger.

The independent-noise diagnostic split each 51-frame response trajectory into four contiguous blocks and evaluated all six non-overlapping two-block selection/evaluation assignments. Molecule-level aggregation preceded 100,000-resample paired bootstrapping. The subsequent resolution protocol was committed at `52827f8` before its power calculation. It retained the empirical centred molecule-effect pattern, shifted its mean to an oracle benefit of 0.30 kcal mol^{-1} and scaled within-molecule variance from 2.5 ps by the inverse production time. A simulation could launch only if power at both budgets and projected dense-reference reliability were each at least 0.80; that gate failed, so no new trajectory was generated.